

Cosas que corregir del artículo

Reseñas que nos dieron:

This paper presents the design, implementation, and preliminary evaluation of a virtual reality-based physical rehabilitation system tailored for university athletes. Developed using the Unity3D Game Engine, the system uses SlimeVR inertial motion units (IMU) to track body movements and provide real-time feedback during shoulder and knee mobility exercises. The study evaluates the system across three dimensions—usability, subjective experience, and functionality—using a group of 30 healthy university students to test the feasibility of wearable sensor-based tracking as an alternative to camera-based systems. #Strengths: + Preliminary results showed exceptionally high usability scores, with interface organization and movement clarity receiving average ratings above 9.5 out of 10. + The survey used for evaluation demonstrated high reliability, with a Cronbach's alpha of 0.96 for the overall rating. + The system includes dynamic goals that progressively adjust based on patient performance, addressing the common issue of boredom in traditional rehabilitation. Weaknesses: - The evaluation was conducted on healthy students simulating exercises rather than patients with actual musculoskeletal injuries. - Limited exercise library: The current project only supports two specific exercises (shoulder and knee), limiting its immediate therapeutic range. - While an instructor provided initial feedback, continuous professional medical supervision was not maintained during all testing sessions. #Major Comments: - The K-Means analysis identified a "Skeptics" group (n=2) who gave significantly lower scores (approximately 7/10). The paper should further investigate whether these lower scores were due to specific body types or technical glitches to ensure the system is inclusive. - The lowest statistical rating was for "Error detection" (8.67). This suggests that while the system tracks repetitions well, it may struggle to identify incorrect form, which is a critical safety requirement for unmonitored rehabilitation. - The paper lacks general feedback from rehabilitation experts on how well the presented system could be integrated into existing pipelines. - Since the paper is below the page limit, the authors should include more implementation details. #Minor Comments: - There are several problems with citations: missing spaces (modularly (Zhou and Hu, 2008)), missing references (repetitions (?)), and separate citations for multiple references (activities (Amin et al., 2025) (Heo et al., 2022)). - The diagram should be included in vector form, not as a bitmap. - Figures 2–6 and 8–10 are not referenced in the text. - Some participants noted that the virtual environment could be more immersive (8.87 rating). - The paper suggests refining user instructions to avoid inconvenience when first operating the program. #Verdict: Based on the above comments, the paper provides a solid foundation for a portable, sensor-based rehabilitation tool that overcomes the technical hurdles of camera-dependent systems. While the high usability scores and robust architecture are promising, the study remains "preliminary" until it is validated in a clinical setting with actual patients.

This paper presents a relevant and practical topic: a Unity-based virtual rehabilitation system using SlimeVR inertial sensors for shoulder and knee exercises. The paper has a clear application focus, and the prototype seems usable and potentially useful as an interactive rehabilitation support tool. However, in its current form, the paper remains too preliminary from a scientific perspective. The paper discusses Kinect-based systems, but there is no direct experimental comparison with camera-based tracking, other IMU systems, or even a no-VR or conventional exercise baseline. This significantly limits the contribution. The paper does not provide enough detail about: - sensor placement and calibration, - motion-processing pipeline, - repetition-detection logic, - error thresholds, - how accuracy/responsiveness were operationalized, - statistical analysis procedure. Also, the experimental protocol was not detailed enough. There are multiple English issues, awkward phrasing, repeated sentences, citation inconsistencies, and formatting problems. To be considered as a conference contribution, the paper should be reframed explicitly as a preliminary usability study or pilot study, not as evidence of rehabilitation efficacy. The methodology section should be expanded with precise technical detail. Please improve the discussion by clearly separating what the study shows from what it does not show. Some example minor issues: - First and second sentences in the abstract should be probably at the end of the abstract? - All Figures could have better resolution. - Fig. 11 has a non-English title.

Resumen de la IA

1. El problema de los "escépticos" (los 2 usuarios que puntuaron bajo)

- Encontraste 2 personas que dieron notas bajas (7/10).
- **Tienes que investigar por qué:** ¿fue por su complexión física? ¿Por algún fallo técnico? ¿Porque no entendieron algo?
- Si no lo analizas, tu sistema parece poco inclusivo o poco fiable.

2. Detección de errores (puntuación más baja: 8.67)

- El sistema cuenta repeticiones bien, pero **no detecta si el ejercicio se hace mal**.
- Eso es un problema de seguridad si alguien lo usara sin supervisión.
- **Debes explicar cómo mejorarías eso** (por ejemplo, añadiendo ángulos límite o alertas).

3. Falta de opinión de expertos reales (médicos o fisioterapeutas)

- Solo tuviste un instructor deportivo al principio.
- **Necesitas feedback de profesionales de rehabilitación** que digan si esto tendría sentido en una clínica o con pacientes reales.

4. Faltan detalles técnicos importantes

El revisor dice que no explicas bien:

- Dónde se colocan los sensores exactamente.
- Cómo se calibran.
- Cómo detectas una repetición correcta (ángulos, umbrales, etc.).
- Cómo mediste la "precisión" o "respuesta" del sistema.
- Qué análisis estadístico usaste (más allá del alpha de Cronbach).

→ ☐ **Tienes que añadir un par de párrafos técnicos con esos detalles.**

5. No comparaste con nada

- Hablas de Kinect, pero **no hiciste una comparación experimental** con Kinect, con otros sensores IMU, ni siquiera con ejercicios normales sin realidad virtual.
- Sin comparación, no demuestras que tu sistema sea mejor ni diferente.

→ ☐ Aunque sea con datos simples, di algo como: *"A diferencia de Kinect, nuestro sistema funcionó bien en condiciones de poca luz"* (si es cierto).

6. Problemas con citas y referencias

- Faltan espacios: modularly (Zhou and Hu, 2008) → debe ser modularly (Zhou & Hu, 2008).
- Hay citas incompletas: repetitions (?) → ahí falta una referencia.

- Citas separadas mal: (Amin et al., 2025) (Heo et al., 2022) → debe ser (Amin et al., 2025; Heo et al., 2022).

7. Figuras

- El diagrama (Figura 1) debe estar en **formato vectorial** (no imagen de mapa de bits).
- Las figuras 2, 3, 4, 5, 6, 8, 9, 10 **no se mencionan en el texto**. Cada figura debe ser citada al menos una vez.

8. Redacción y formato

- El título de la Figura 11 está en **otro idioma** (no inglés).
- Hay frases repetidas (por ejemplo, en la sección 2.1 se repite un párrafo entero).
- El abstract está mal estructurado: las dos primeras frases deberían ir al final del resumen.

9. Entorno virtual poco inmersivo

- Los usuarios puntuaron 8.87 la "inmersión".
- Sugiere mejorar el entorno visual para que sea más atractivo.

10. Instrucciones iniciales

- Algunos usuarios tuvieron problemas al empezar.
- Mejora la pantalla de instrucciones para que sea más clara.

Área	Acción concreta
Escépticos	Analiza por qué esos 2 usuarios puntuaron bajo.
Detección de errores	Propón cómo mejorar la detección de mala ejecución.
Expertos	Incluye opinión de un fisioterapeuta o médico.
Detalles técnicos	Explica sensores, calibración, detección de repeticiones, umbrales.
Comparación	Aunque sea teórica, compara con Kinect o con ejercicio sin VR.
Citas	Revisa espacios, citas incompletas y formato.
Figuras	Cita cada figura en el texto.
Texto repetido	Elimina párrafos duplicados.
Inmersión	Menciona que mejorarás el entorno visual.
Título de figura	Traduce al inglés.

Todo para antes del 15 de abril